

%% Objective 2 : LMS

%%%

```
w = zeros(L,1);
y_lms = zeros(N,1);
e_lms = zeros(N,1);
for n=L:N
    xvec = x(n:-1:n-L+1);
    y_lms(n) = w'*xvec;
    e_lms(n) = d(n)-y_lms(n);
    w = w + mu*e_lms(n)*xvec;
```

end

figure;

plot(e_lms,'LineWidth',1);

grid on;

title('LMS Output');

xlabel('Samples');

ylabel('Amplitude');

%%%

%% Objective 3 : NLMS

%%%

```
w = zeros(L,1);
y_nlms = zeros(N,1);
e_nlms = zeros(N,1);
for n=L:N
    xvec = x(n:-1:n-L+1);
    y_nlms(n)=w'*xvec;
    e_nlms(n)=d(n)-y_nlms(n);
    w=w+(mu_nlms/(delta+xvec'*xvec))*e_nlms(n)*xvec;
```

end

figure;

plot(e_nlms,'LineWidth',1);

grid on;

title('NLMS Output');

xlabel('Samples');

ylabel('Amplitude');

%%%

%% Objective 4 : MSE Comparison

%%%

MSE_LMS=zeros(N,1);

```

MSE_NLMS=zeros(N,1);
for k=1:N
    MSE_LMS(k)=mean(e_lms(1:k).^2);
    MSE_NLMS(k)=mean(e_nlms(1:k).^2);
end
figure;
semilogy(MSE_LMS,'r','LineWidth',1.2);
hold on;
semilogy(MSE_NLMS,'b','LineWidth',1.2);
grid on;
legend('LMS','NLMS');
title('MSE Comparison');
xlabel('Iterations');
ylabel('Mean Square Error');

```

OUTPUT:





